

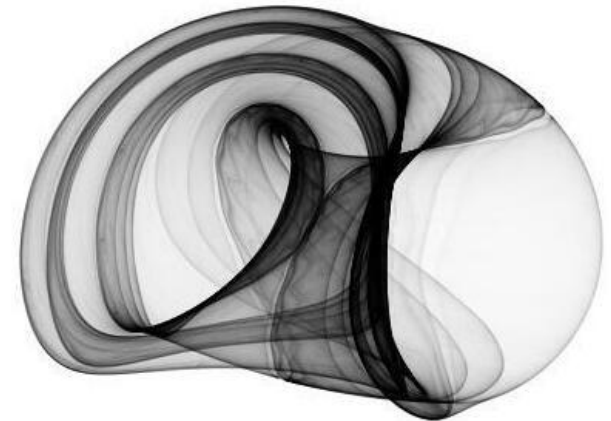
Out-of-time-order correlators (OTOC) and quantum chaos in random quantum circuits

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Quantum chaos and nonlinear dynamics

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Out-of-time-order correlators (OTOC)

the out-of-time-order correlator (**OTOC**)¹, describes the averaged evolution of quantum Heisenberg operators in time.

Black-holes were shown to exhibit the early exponential time-growth of the OTOC expected in classically chaotic systems, leading to the popular terminologies of “**scrambling**”, “**quantum butterfly effect**”, and “**quantum Lyapunov exponent**”.

¹ Larkin, A. I., & Ovchinnikov, Y. N. (1969). Quasiclassical method in the theory of superconductivity. *Sov Phys JETP*, 28(6), 1200-1205.

Out-of-time-order correlators (OTOC)

Moreover, we describe the OTOC as an established invaluable tool to comprehend and characterize **complex systems**, especially when undergoing a transition from regular to chaotic regime and when no classical analogue exists to guide our intuition.²

$$\mathcal{C}_{V\hat{W}}(t) = \left\langle [\hat{W}(t), \hat{V}]^\dagger [\hat{W}(t), \hat{V}] \right\rangle$$

The sub-index t stands for the time-evolution of the operator $\hat{O}(t)$ under the Hamiltonian $\hat{H}(t)$ in the Heisenberg representation, *i. e* $\hat{O}(t) = \hat{U}^\dagger(t) \hat{O} \hat{U}(t)$ with $\hat{U}(t) = e^{-i\hat{H}t/\hbar}$

² García-Mata, I., Jalabert, R. A., & Wisniacki, D. A. (2022). Out-of-time-order correlators and quantum chaos. *arXiv preprint arXiv:2209.07965*.

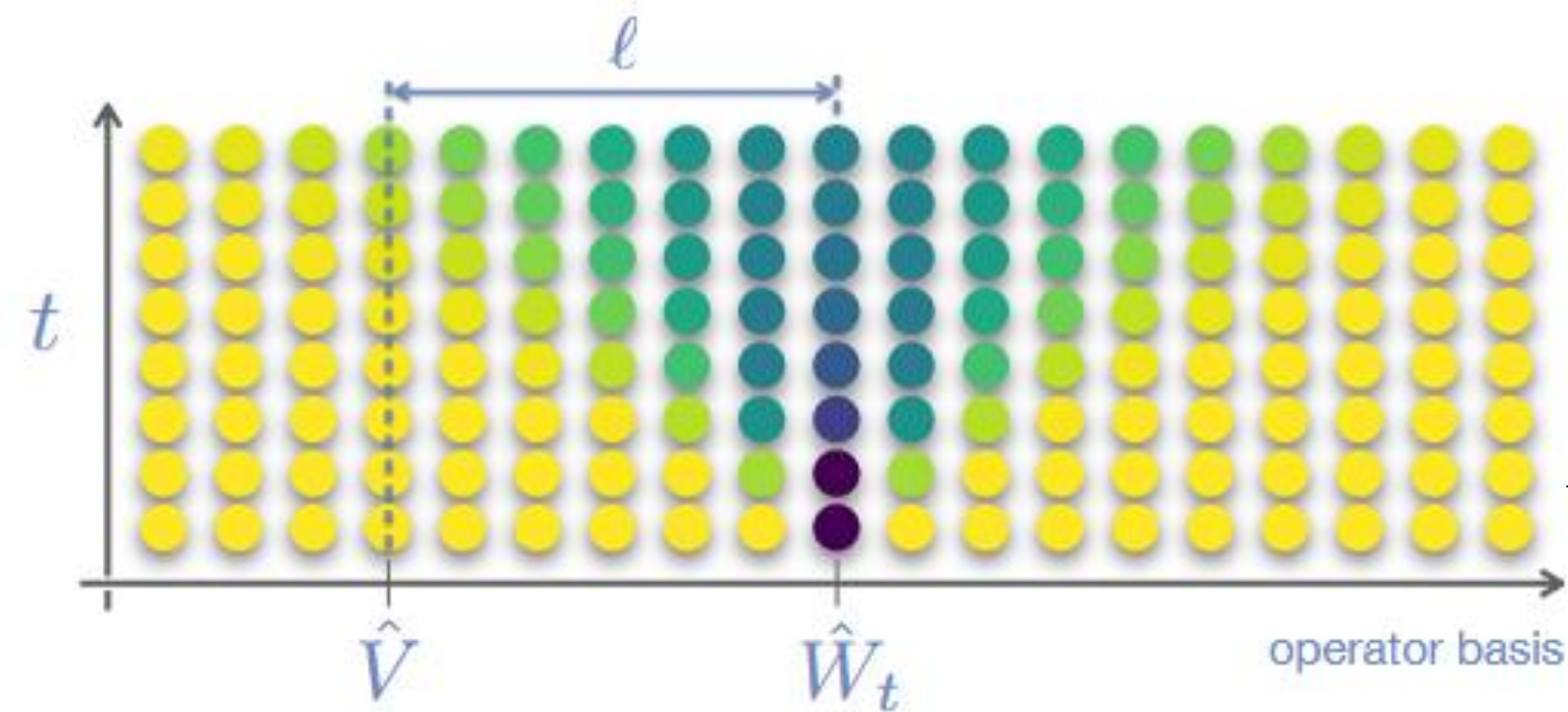
Out-of-time-order correlators (OTOC)

The previous definition holds for an arbitrary pair of operators. However, with the goal of characterizing **physical observables** and in the search of possible universal behavior, the definition of the OTOC is typically restricted to the case in which operators $\hat{V}$ and $\hat{W}$ **are Hermitian**, where:



$$\mathcal{C}_{V\hat{W}}(t) = - \left\langle [\hat{W}(t), \hat{V}]^2 \right\rangle$$

Out-of-time-order correlators (OTOC)



The Baker-Campbell-Hausdorff formula for the expansion of $\hat{W}(t)$ yields the nested commutator expression

$$\hat{W}(t) = \sum_{k=0}^{\infty} \frac{(it)^k}{k!} \underbrace{[\hat{H}, \dots [\hat{H}, \hat{W}] \dots]}_k$$

² García-Mata, I., Jalabert, R. A., & Wisniacki, D. A. (2022). Out-of-time-order correlators and quantum chaos. *arXiv preprint arXiv:2209.07965*.

Out-of-time-order correlators (OTOC)

In the quasi-classical limit of $\hbar \rightarrow 0$, given the well-known relation

$$\lim_{\hbar \rightarrow 0} \frac{1}{i\hbar} [\hat{W}, \hat{V}] = \{W, V\},$$

making the choice $\hat{W} = \hat{X}$ and $\hat{V} = \hat{P}_X$, the quasi-classical limit of the commutator $[\hat{X}(t), \hat{P}_X]$ can be expressed as;

$$\{X(t), P_X(0)\} = \frac{\partial X(t)}{\partial X(0)} \cdot \underbrace{\frac{\partial P_X(0)}{\partial P_X(0)}}_{=1} - \frac{\partial X(t)}{\partial P_X(0)} \cdot \underbrace{\frac{\partial P_X(0)}{\partial X(0)}}_{=0} = \frac{\partial X(t)}{\partial X(0)}$$



$$[X(t), P_X(0)] \approx i\hbar \frac{\partial X(t)}{\partial X(0)}$$

² García-Mata, I., Jalabert, R. A., & Wisniacki, D. A. (2022). Out-of-time-order correlators and quantum chaos. *arXiv preprint arXiv:2209.07965*.

OTOC and Lyapunov exponent

So, the quasi-classic OTOC is;

$$C_{\hat{P}_X \hat{X}}^{qc}(t) = -\langle [\hat{X}(t), \hat{P}_X(0)]^2 \rangle \approx \hbar^2 \left\langle \left(\frac{\partial X(t)}{\partial X(0)} \right)^2 \right\rangle \rightarrow$$

In a classically chaotic system,
this derivative grows
exponentially over time:

$$\frac{\partial X(t)}{\partial X(0)} \sim e^{\lambda t}$$

Let be $\delta \vec{X}(0)$ a small disturbance to the initial state of a trajectory $\vec{X}(t)$, The **Lyapunov exponent** λ is defined classically as:

$$\lambda = \lim_{t \rightarrow \infty} \lim_{\|\delta \vec{x}(0)\| \rightarrow 0} \frac{1}{t} \ln \frac{\|\delta \vec{x}(t)\|}{\|\delta \vec{x}(0)\|}$$

- ☐ $\lambda > 0$: Chaos sensitive to initial conditions.
- ☐ $\lambda < 0$: Attraction.
- ☐ $\lambda = 0$: Marginal instability.

² García-Mata, I., Jalabert, R. A., & Wisniacki, D. A. (2022). Out-of-time-order correlators and quantum chaos. *arXiv preprint arXiv:2209.07965*.

OTOC and Lyapunov exponent

If the classical dynamics is fully chaotic, the exponential sensitivity with respect to the initial condition translates into $\partial X(t)/\partial X(0) \sim \exp(\lambda t)$, where λ is the Lyapunov exponent. Therefore, the **quasi-classical limit of the OTOC** is given by

$$\mathcal{C}_{\hat{P}_X \hat{X}}(t) \sim h^2 \exp(2\lambda t).$$

This relationship is particularly interesting, since it establishes the signature of the Lyapunov exponent, which is a purely classical quantity, in the OTOC, which is a genuine quantum object.

OTOC and Lyapunov exponent

A bound on chaos

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Abstract

We conjecture a sharp bound on the rate of growth of chaos in thermal quantum systems with a large number of degrees of freedom. Chaos can be diagnosed using an out-of-time-order correlation function closely related to the commutator of operators separated in time. We conjecture that the influence of chaos on this correlator can develop no faster than exponentially, with Lyapunov exponent $\lambda_L \leq 2\pi k_B T / \hbar$. We give a precise mathematical argument, based on plausible physical assumptions, establishing this conjecture.

It was shown by Maldacena–Shenker–Stanford³, that, under certain conditions (unitarity, strong coupling, finite temperature), the exponential growth of the OTOC is limited by:

$$\lambda_L \leq \frac{2\pi k_B T}{\hbar}$$

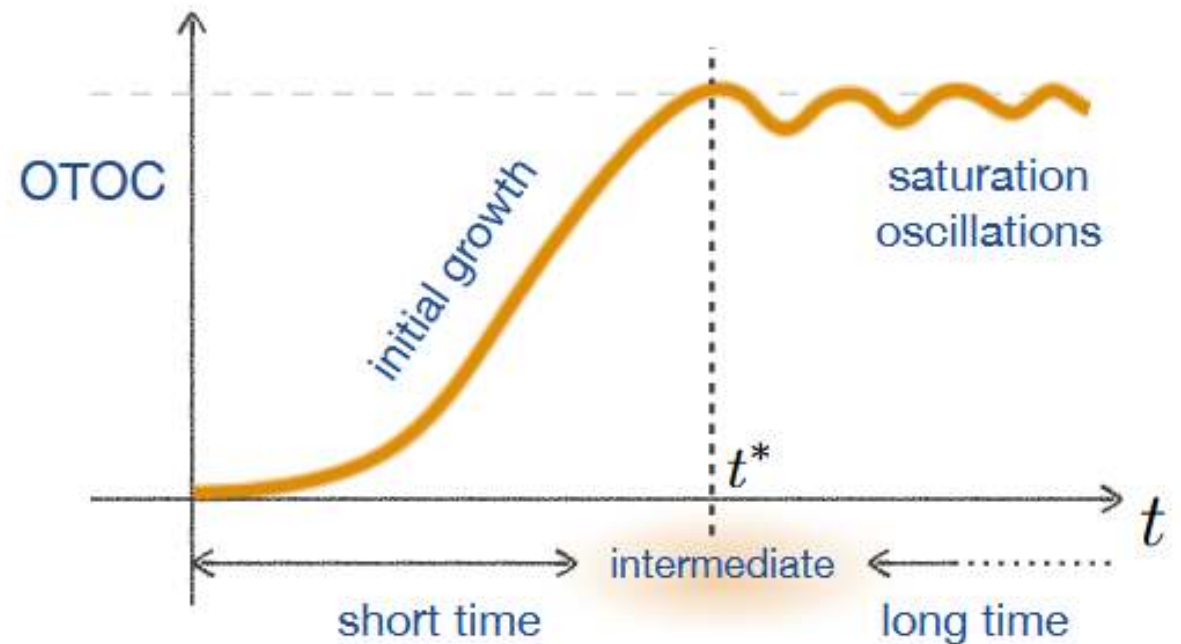
³ Maldacena, J., Shenker, S. H., & Stanford, D. (2016). A bound on chaos. *Journal of High Energy Physics*, 2016(8), 1-17.

Time regimes

While the quasi-classical form of the OTOC points to the emergence of classical features in quantum systems, it is clear that such kind of universal behavior can only appear in a **well-defined class of systems**, and in a **particular time-window**.

$$\mathcal{C}_{\hat{P}_X \hat{X}}(t) \sim h^2 \exp(2\lambda t).$$

the OTOC description can in general be divided in three time-regimes



Short times

In the short-time regime, the OTOC captures the **initial growth** associated with the spread of the Heisenberg-evolved operator $\hat{W}(t)$, under the system's Hamiltonian dynamics.

Conditions for Exponential Growth:

- ❑ Quantum systems with **well-defined classical chaotic limits**, such as: Quantum maps, Billiards, Kicked rotor (in the chaotic regime).
- ❑ Use of **well-defined operators** $\hat{X}$, $\hat{P}$ or their discrete analogs in finite Hilbert spaces

² García-Mata, I., Jalabert, R. A., & Wisniacki, D. A. (2022). Out-of-time-order correlators and quantum chaos. *arXiv preprint arXiv:2209.07965*.

Short times

In the short-time regime, the OTOC captures the **initial growth** associated with the spread of the Heisenberg-evolved operator $\hat{W}(t)$, under the system's Hamiltonian dynamics.

Notable Exceptions:

- ❑ Systems **without a classical analogue**, such as spin-1/2 chains, typically show **power-law growth**.
- ❑ Systems where the operator $\hat{W}$ is localized near a **phase-space saddle point** may also exhibit exponential growth without genuine chaos (no saturation or scrambling).

² García-Mata, I., Jalabert, R. A., & Wisniacki, D. A. (2022). Out-of-time-order correlators and quantum chaos. *arXiv preprint arXiv:2209.07965*.

Intermediate times

After the initial growth, around the **scrambling time** t^* , the OTOC settles to an approximately constant value, with small oscillations in some cases, it can be understood as the time between t^* and complete relaxation.



For systems with a classical analogue, t^* corresponds to the **Ehrenfest time**:

$$t_E \sim \frac{1}{\lambda} \ln \left(\frac{a}{\hbar} \right)$$

² García-Mata, I., Jalabert, R. A., & Wisniacki, D. A. (2022). Out-of-time-order correlators and quantum chaos. *arXiv preprint arXiv:2209.07965*.

Long times

In the case of strongly chaotic systems, the relaxation after the scrambling time leads to a **constant value of the OTOC**, with small and irregular fluctuations.

The **saturation value** depends on:

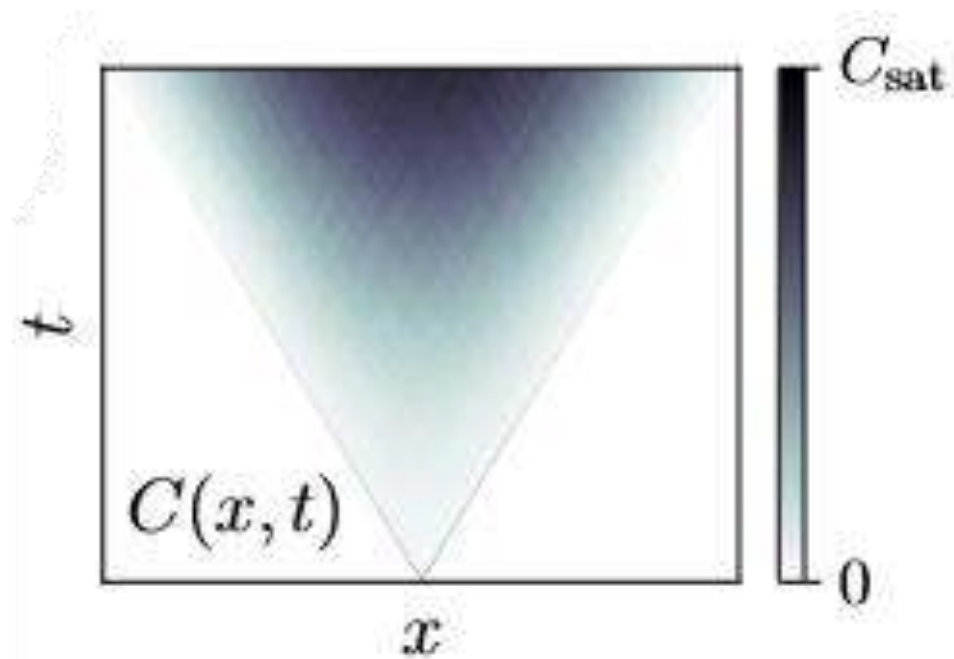
- ❑ The **system size** N .
- ❑ The **temperature** T .
- ❑ The choice of operators $\hat{V}$ and $\hat{W}$.

For $\hat{W} = \hat{X}$ and $\hat{V} = \hat{P}_X$, in fully chaotic many-body systems:

$$C_\infty \sim \hbar^2 T N$$

Many-Body Quantum Chaos

In quantum many-body systems, chaos manifests as the rapid spread of initially localized quantum information, a process known as **scrambling**. This phenomenon is quantified by the **OTOC**; can be understood how **non-commuting** initially local operators evolve and **affect** distant parts of the system under Heisenberg time evolution.



The spatial spread of operators is governed by the **butterfly velocity** v_B , which determines the size of the operator's light cone.

$$C_{V\hat{W}}(t) = - \left\langle [\hat{W}(t), \hat{V}]^2 \right\rangle$$

² García-Mata, I., Jalabert, R. A., & Wisniacki, D. A. (2022). Out-of-time-order correlators and quantum chaos. *arXiv preprint arXiv:2209.07965*.

Many-Body Quantum Chaos

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Letter

Scrambling and Lyapunov exponent in spatially extended systems

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Scrambling of information in a quantum many-body system, quantified by the out-of-time-ordered correlator (OTOC), is a key manifestation of quantum chaos. A regime of exponential growth in the OTOC, characterized by a Lyapunov exponent, has so far mostly been observed in systems with a high-dimensional local Hilbert space and in weakly-coupled systems. Here, we propose a general criterion for the existence of a well-defined regime of exponential growth of the OTOC in spatially extended systems with local interactions. In such systems, we show that a parametrically long period of exponential growth requires the butterfly velocity to be much larger than the Lyapunov exponent times a microscopic length scale, such as the lattice spacing. As an explicit example, we study a random unitary circuit with tunable interactions. In this model, we show that in the weakly interacting limit, the above criterion is satisfied, and there is a prolonged window of exponential growth. Our results are based on numerical simulations of both Clifford and universal random circuits supported by an analytical treatment.

Scrambling time in systems with local interactions

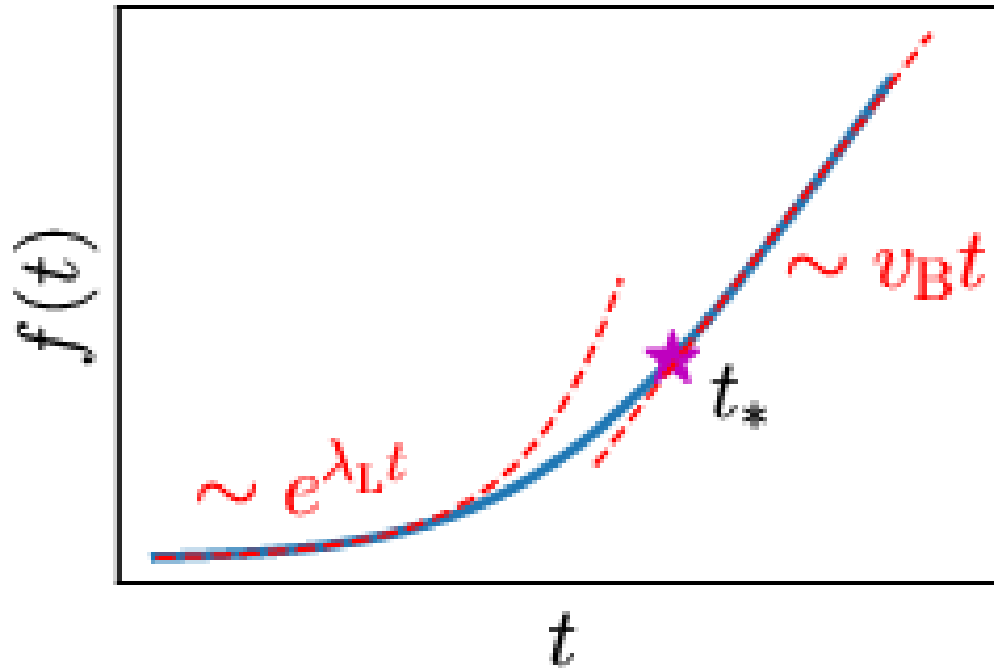
To better characterize the global scrambling behavior, the integrated OTOC (**iOTOC**) is introduced as,

$$f(t) = \sum_j C_{i,j}(t) = - \sum_j \langle [W_i(t), V_j]^2 \rangle,$$

which measures the **total non-commutativity** of a time-evolved operator $\hat{W}_i(t)$ with all other local operators $\hat{V}_j$ across the system.

Scrambling time in systems with local interactions

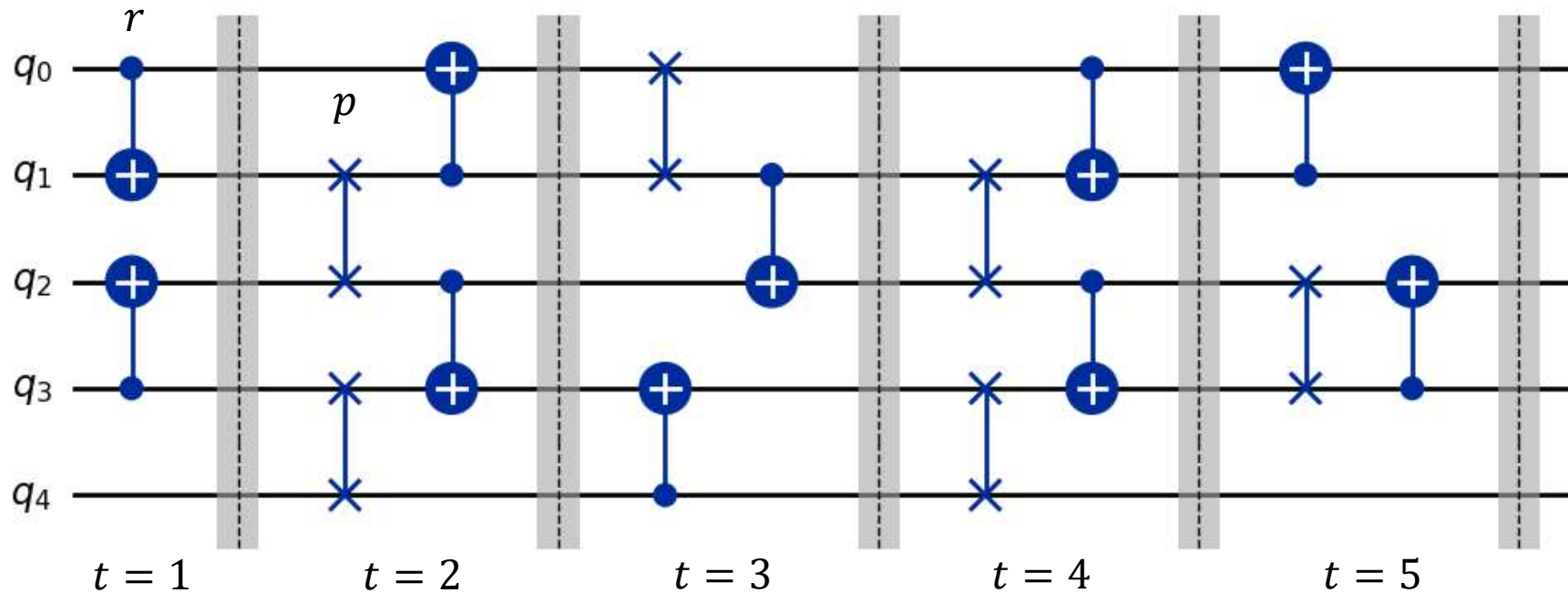
At early times, we expect the iOTOC to exhibit an exponential growth characterized by a **Lyapunov exponent**, while at late times, when the OTOC in the bulk of the system reaches its saturation value C_{sat} , the iOTOC crosses over to a **linear growth** regime (due to the linear growth of the size of the light cone).



$$t^* \sim \frac{1}{\lambda_L} \log \left(\frac{C_{sat} v_B}{\lambda_L} \right)$$

Random Clifford circuit model.

The authors study a random unitary circuit model with **tunable interaction** strength. This model consists of a **one-dimensional chain of qubits**.



At every odd (even) time step, **SWAP** gates are applied probabilistically (with probability p) to odd (even) bonds, allowing for ballistic or diffusive **propagation of operators**.

Additionally, **CNOT** gates are applied to a fraction r of the bonds. These CNOT gates generate **interactions** that cause the local operators to grow in support, leading to scrambling.

QUANTUM GATES

➔ **SWAP gate:** It is a quantum operation that exchanges the states of two qubits, it is a unitary gate and can be defined as;

$$\text{SWAP } |a\rangle|b\rangle = |b\rangle|a\rangle \quad \longrightarrow \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \longrightarrow \quad \begin{array}{c} \text{---} \times \text{---} \\ | \\ \times \text{---} \text{---} \end{array}$$

➔ **CNOT gate:** (Controlled-Not) is a two-qubits gate, one for control and the other for target. Its action is:

$$\text{CNOT}|c\rangle|t\rangle = |c\rangle|t \oplus c\rangle \quad \longrightarrow \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad \longrightarrow \quad \begin{array}{c} \text{---} \bullet \text{---} \\ | \\ \bigoplus \text{---} \end{array}$$

Random Clifford circuit model.

REMARK

The model is studied using **Clifford circuits**, which allow for efficient **classical simulation**. In the absence of CNOT gates ($r = 0$), operators propagate purely via SWAP gates, either **ballistically** ($p = 1$) or **diffusively** ($p < 1$).

When CNOT gates are introduced ($r > 0$), they act as **scattering events** that increase operator complexity and lead to a well-defined scrambling time and Lyapunov exponent.

Note that the circuit consists of **Clifford gates** only.

iOTOC Calculation

For the circuit structure described above, the operator $\hat{\sigma}^\alpha_i(t)$ is considered as $\hat{W}_i(t)$, with $\alpha = x, y, z$, Pauli operators, and $\hat{\sigma}^\beta_j$ as $\hat{V}_j$.

- The **action of a CNOT gate** on operators (when the first qubit is the control qubit and the second one is the target) is given by (for concreteness focus on the evolution of a $\hat{\sigma}^z$ operator):

$$\sigma^z \otimes 1 \rightarrow \sigma^z \otimes 1, 1 \otimes \sigma^z \rightarrow \sigma^z \otimes \sigma^z, \sigma^z \otimes \sigma^z \rightarrow 1 \otimes \sigma^z$$

- The corresponding operator string at times $t > 0$ **consists only** of $\hat{\sigma}^z$ and identity operators.

iOTOC Calculation

For the circuit structure described above, the operator $\hat{\sigma}_i^\alpha(t)$ is considered as $\hat{W}_i(t)$, with $\alpha = x, y, z$, Pauli operators, and $\hat{\sigma}_j^\beta$ as $\hat{V}_j$.

regarding

$$f(t) = \sum_j C_{i,j}(t) = - \sum_j \langle [\sigma_i^\alpha(t), \sigma_j^\beta]^2 \rangle$$

- Writing the operator string corresponding to $\hat{\sigma}_i^\alpha(t)$ explicitly as $\bigotimes_k \hat{\sigma}_k^{\alpha_k(t)}$, the **commutator** takes the form;

$$[\sigma_i^\alpha(t), \sigma_j^\beta] = \left(\bigotimes_{k \neq j} \sigma_k^{\alpha_k(t)} \right) \otimes [\sigma_j^{\alpha_j(t)}, \sigma_j^\beta]$$

$$\hookrightarrow [\sigma_i^\alpha(t), \sigma_j^\beta]^2 = 4(\delta_{\beta, \alpha_j(t)} - 1) \mathbb{I}$$

iOTOC Calculation

Performing the summation over j , with $\hat{W}_i = \hat{\sigma}_i^\alpha / \sqrt{2}$ and $\hat{V}_j = \hat{\sigma}_j^\beta / \sqrt{2}$, we find,

$$f(t) = \sum_j (1 - \delta_{\beta, \alpha_j(t)}) (1 - \delta_{\alpha_j(t), 0})$$

i. e. the number of (non-identity) Pauli operators in $\hat{\sigma}^\alpha_i(t)$ which are different from $\hat{\sigma}^\beta$.

Here is considered the OTOC between $\hat{\sigma}_i^z(t)$ and $\hat{\sigma}_j^x$, so that $f(t)$ amounts to the number of $\hat{\sigma}^z$ operators in the string $\hat{\sigma}_i^z(t)$ at time t .

Master Equation

for the integrated OTOC $f(t)$ in the limit of weak interactions ($r \ll 1$) and nearly ballistic propagation ($1 - p \ll 1$). In this dilute regime, scattering events due to CNOT gates can be treated as uncorrelated, analogous to ***the molecular chaos*** hypothesis in *kinetic theory*,

$$\frac{df}{dt} = r f(t) \left(1 - \frac{f(t)}{v_B t} \right),$$

The full solution is given by:

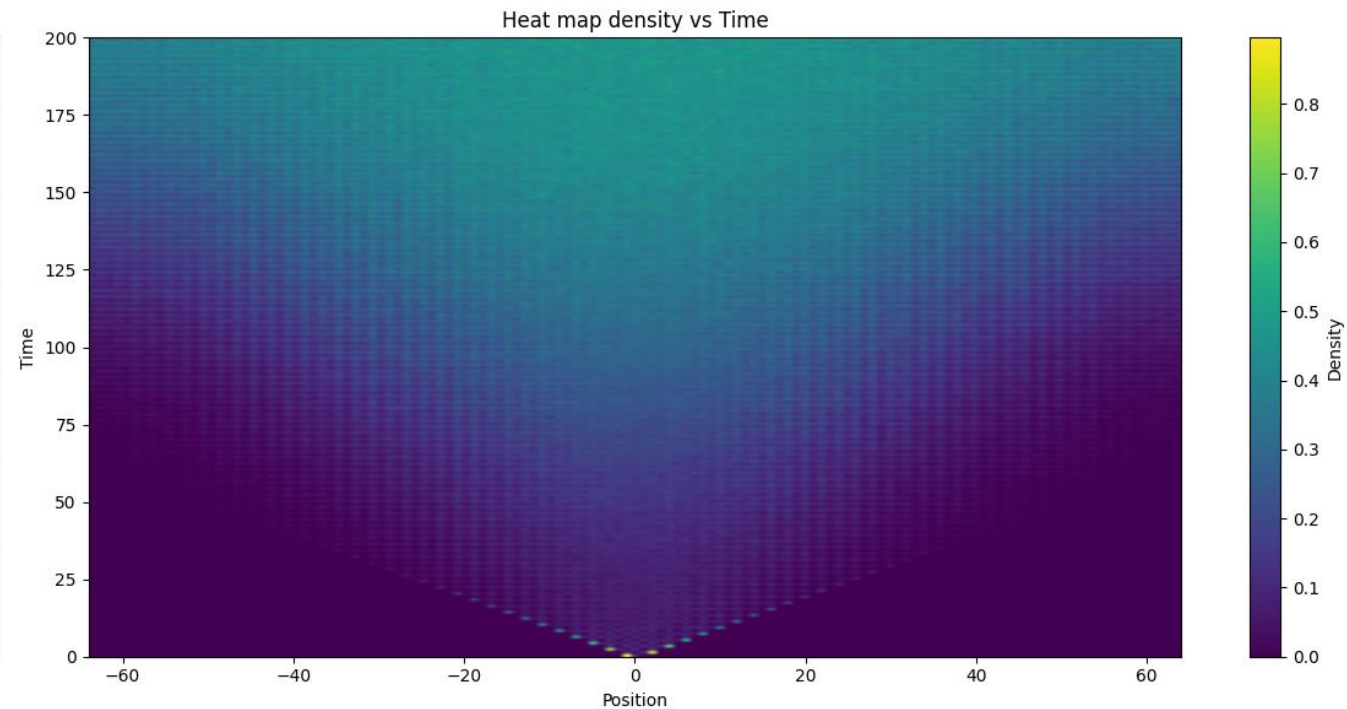
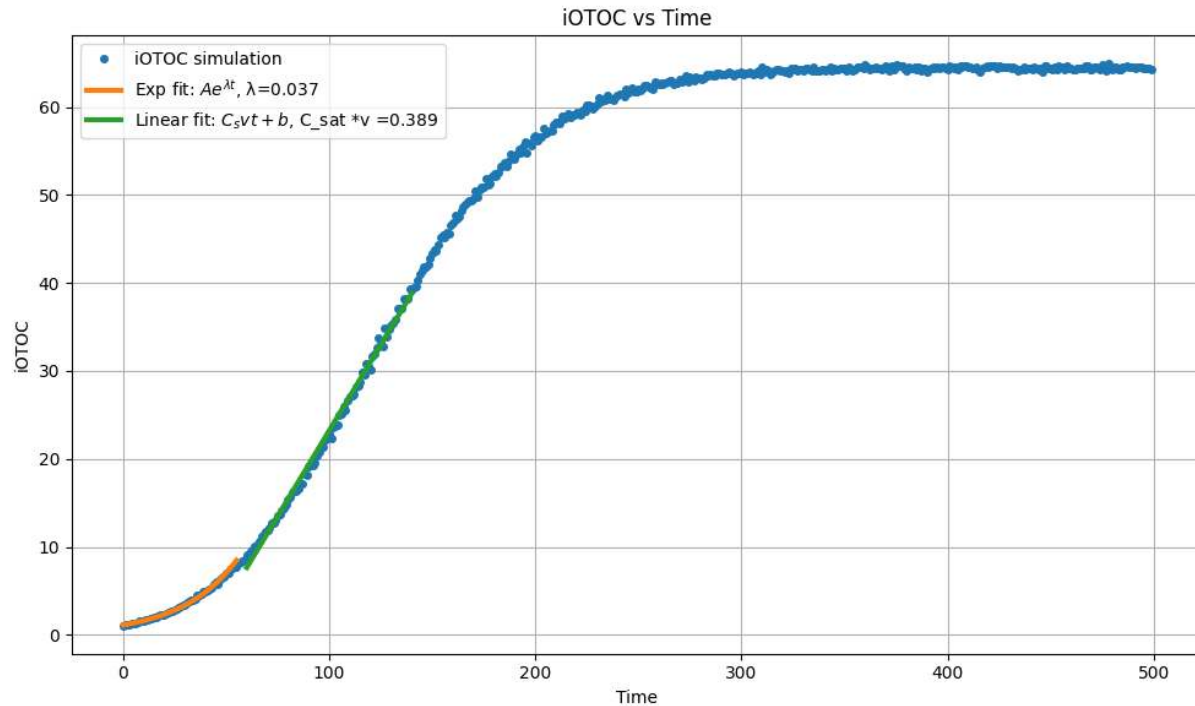
$$f(t) = \frac{g_0 e^{rt}}{1 + g_0 \frac{r}{v_B} [\text{Ei}(rt) - \text{Ei}(1)]}$$

Results

A simulation is developed in Python using **Qiskit**, to explore the behavior of iOTOC.

For the following results, the number of qubits was fixed to 129.

$$p = 0.9, r = 0.05$$

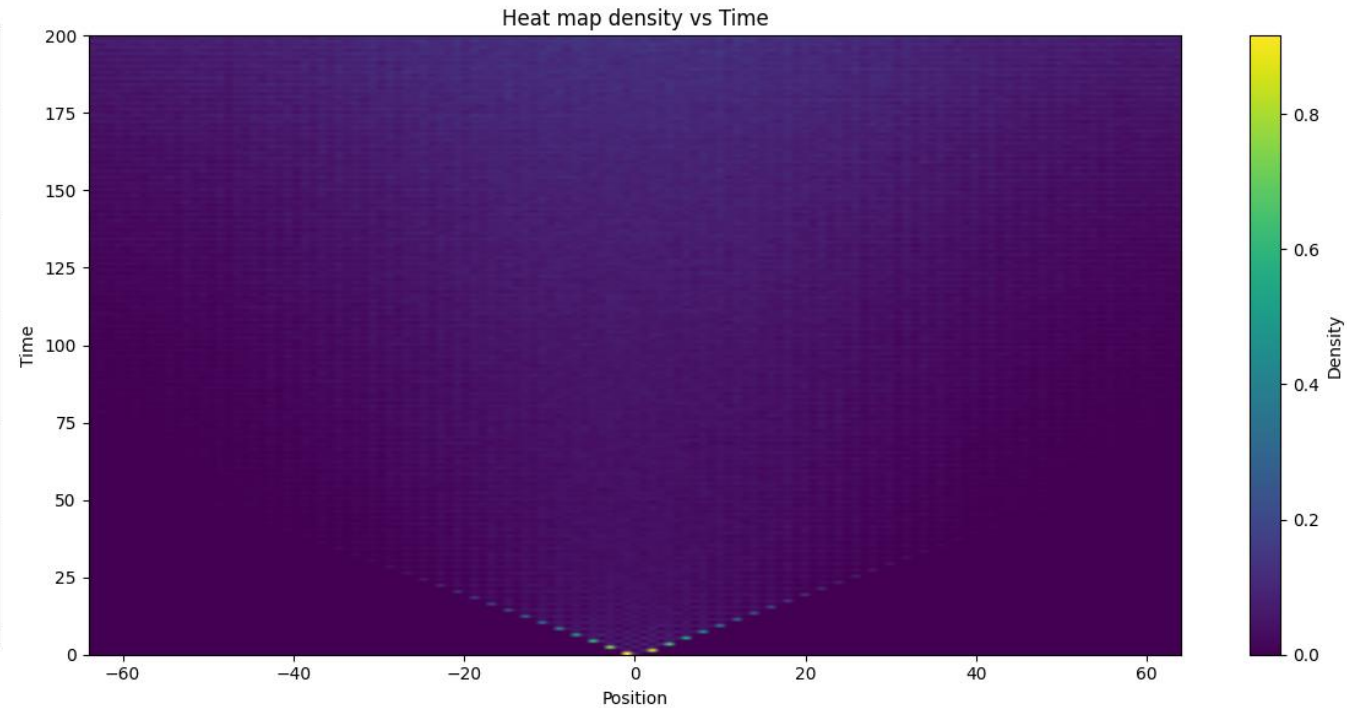
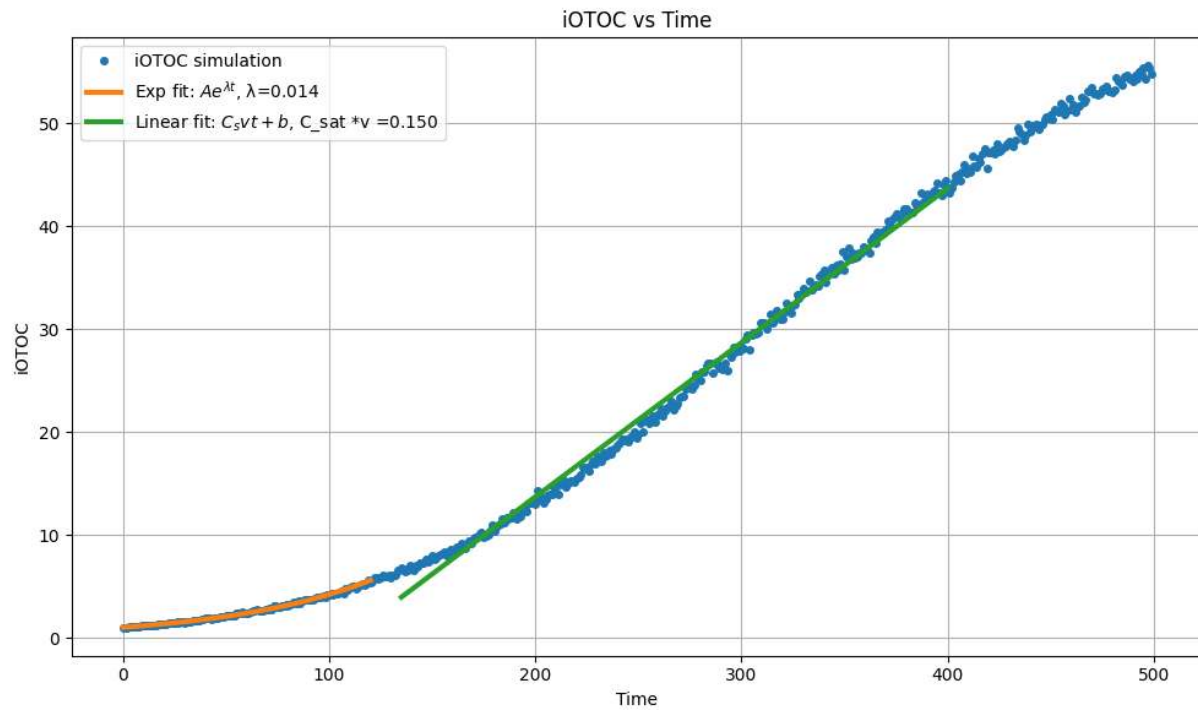


$$\lambda = 0.03689, C_{sat} = 64, v_B = 0.971$$

Results

For the following results, the number of qubits was fixed to 129.

$$p = 0.9, r = 0.02$$

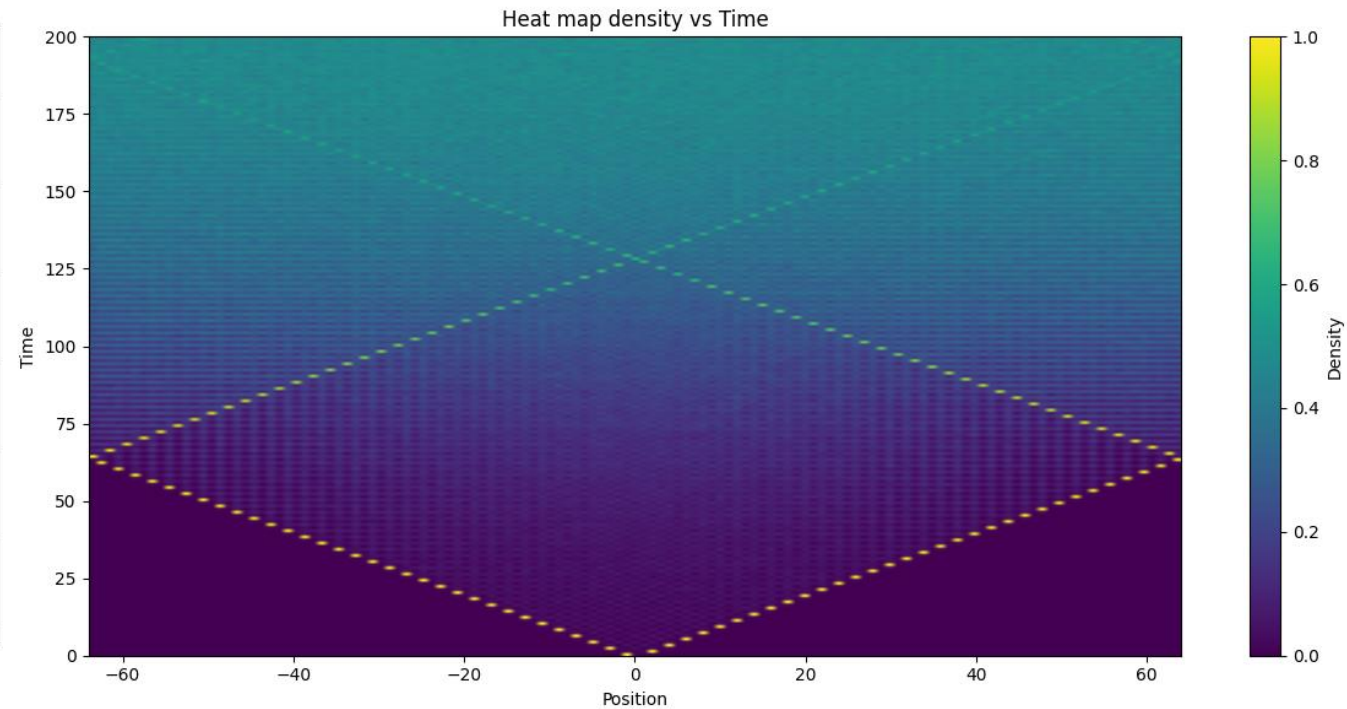
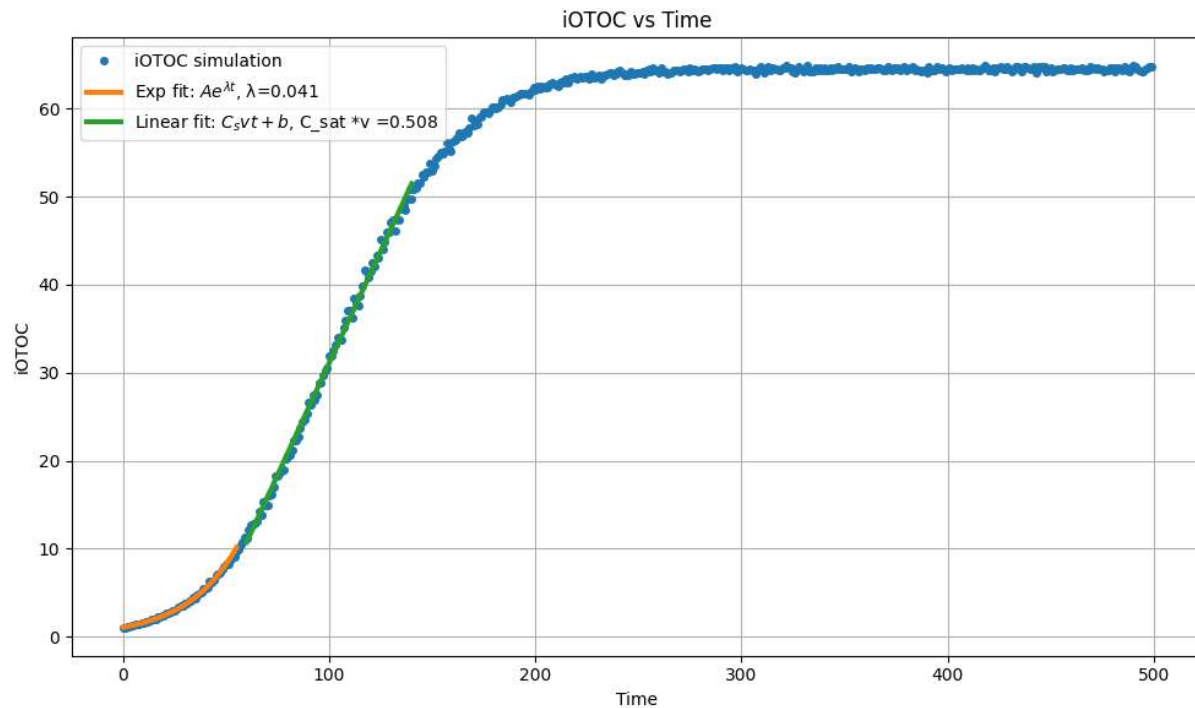


$$\lambda = 0.01403, C_{sat} = 64, v_B = 0.954$$

Results

For the following results, the number of qubits was fixed to 129.

$$p = 1, r = 0.05$$

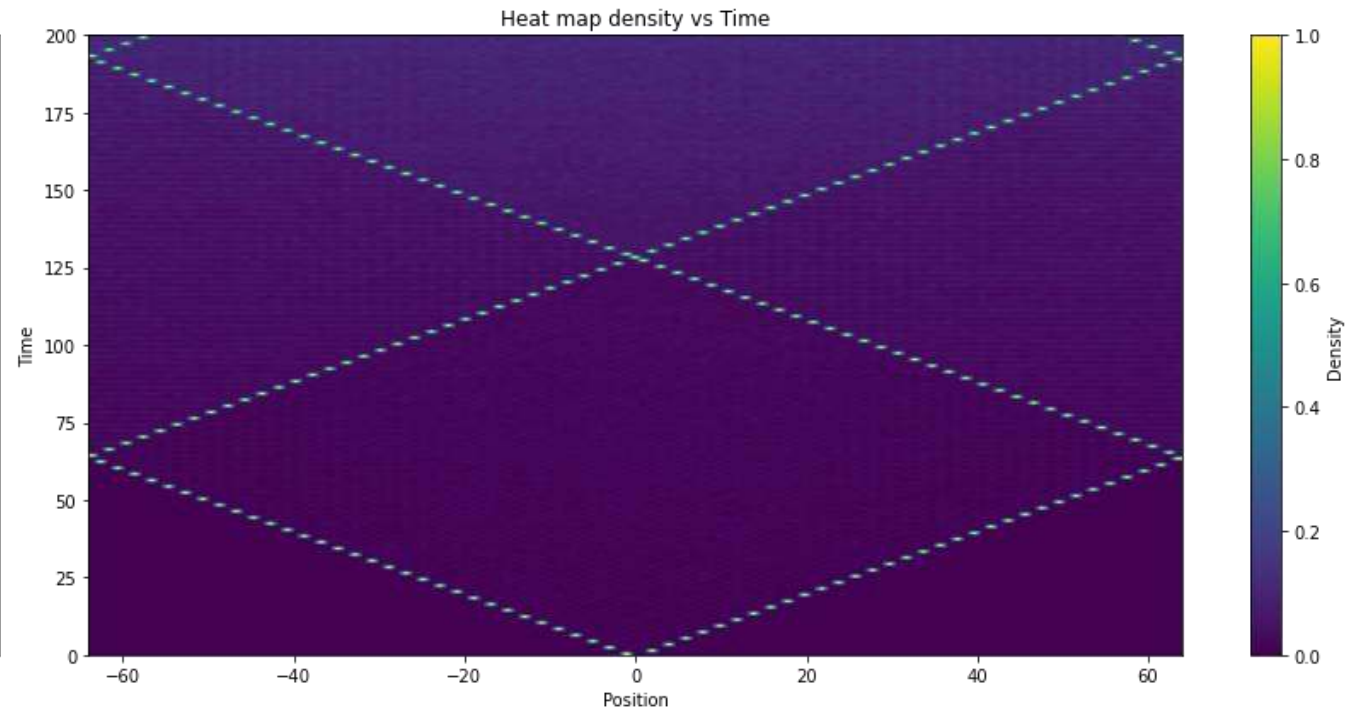
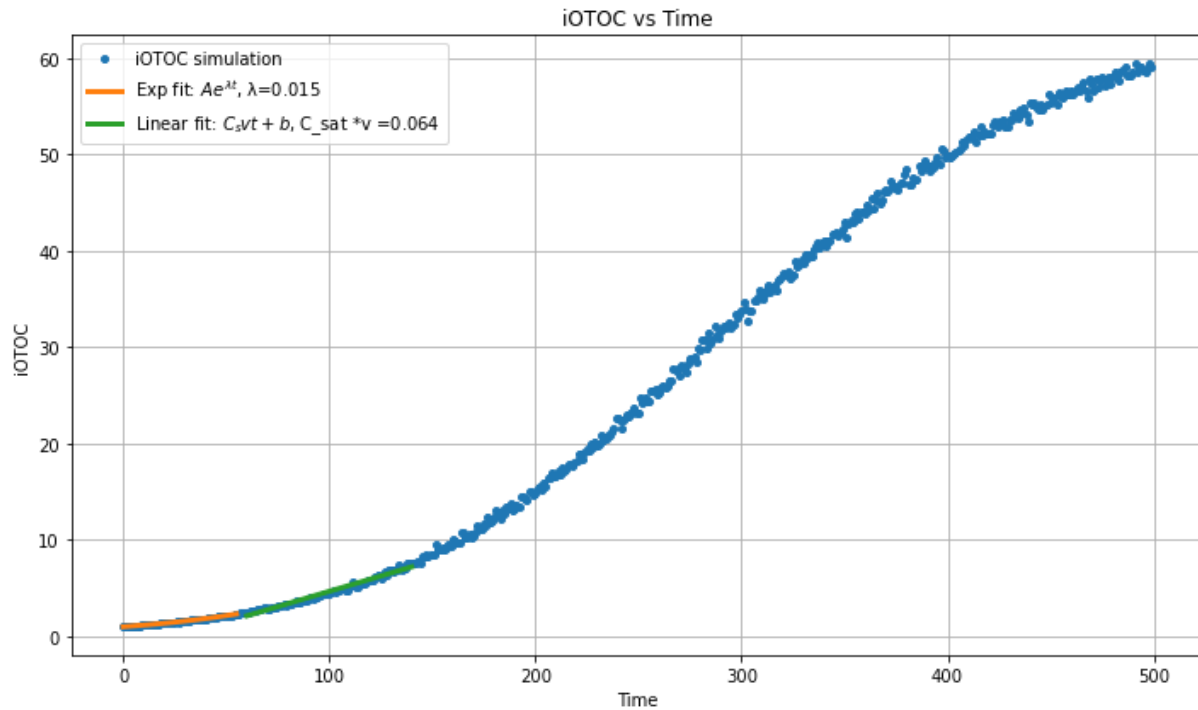


$$\lambda = 0.04080, C_{sat} = 64, v_B = 1.0$$

Results

For the following results, the number of qubits was fixed to 129.

$$p = 1, r = 0.02$$

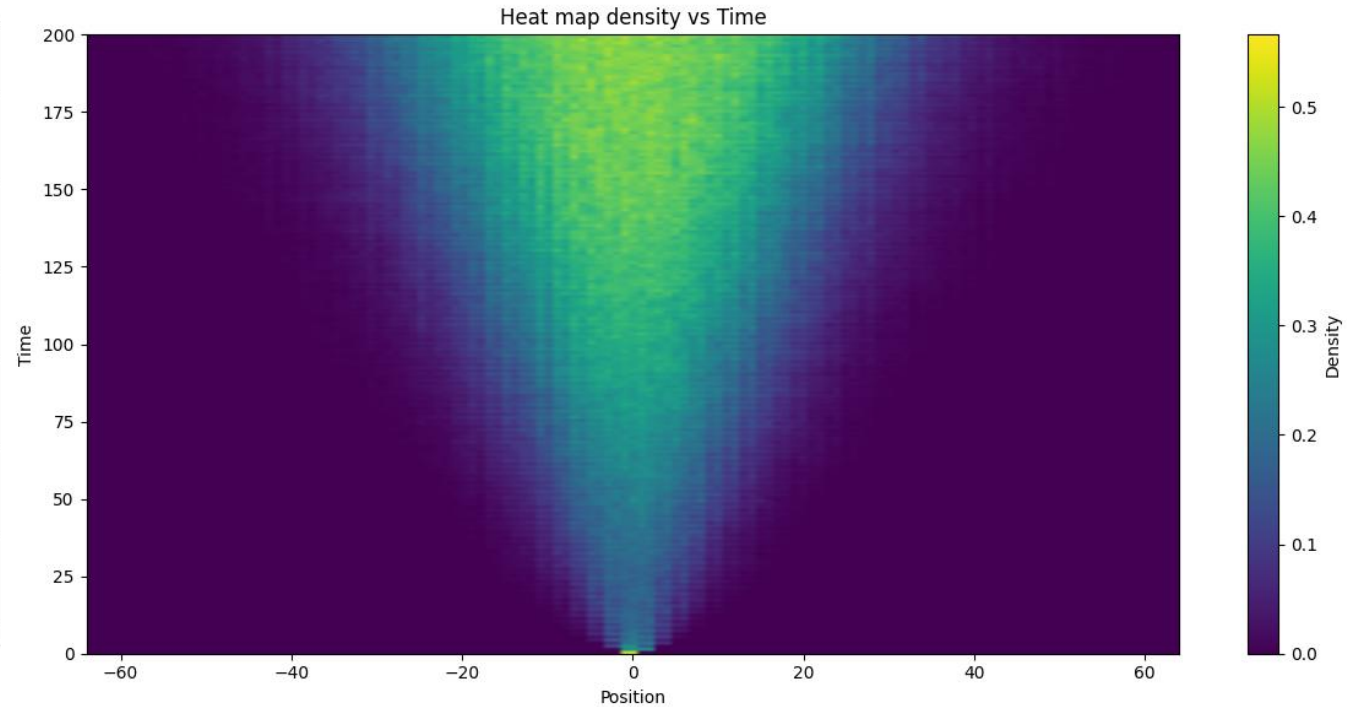
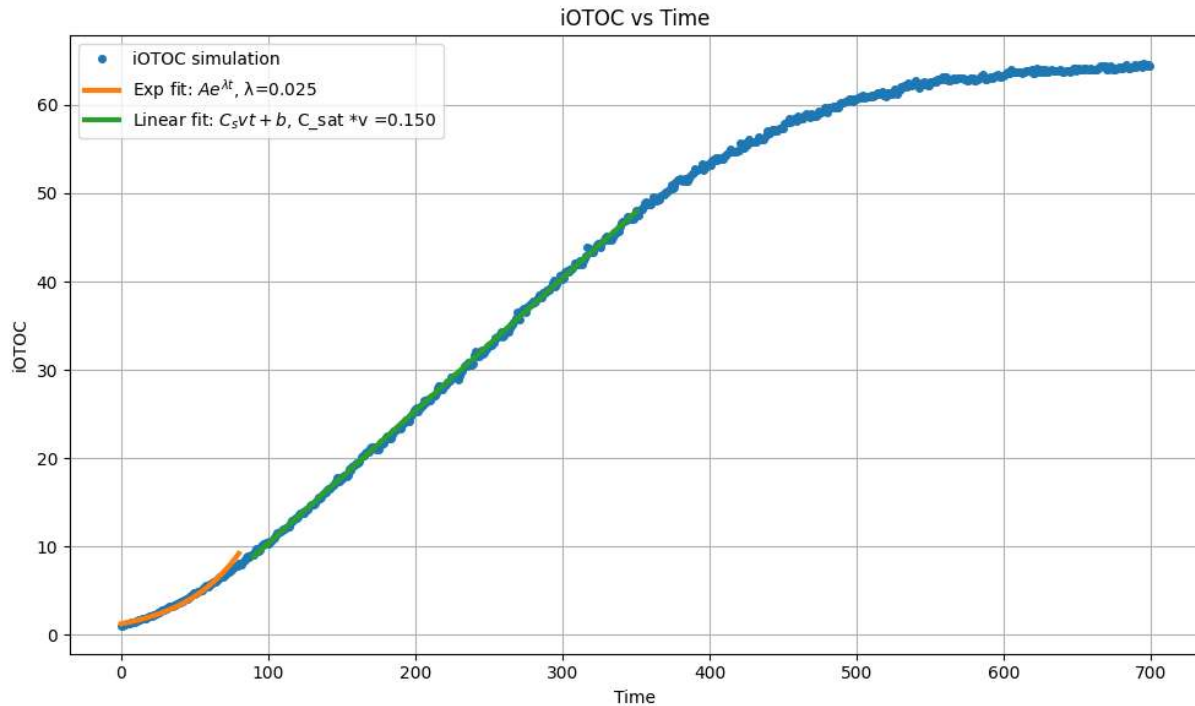


$$\lambda = 0.04080, C_{sat} = 64, v_B = 1.0$$

Results

For the following results, the number of qubits was fixed to 129.

$$p = 0.5, r = 0.05$$



$$\lambda = 0.02484, C_{sat} = 64, v_B = 0.277$$

Quantum Chaos is Quantum

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It is well known that a quantum circuit on N qubits composed of Clifford gates with the addition of k non Clifford gates can be simulated on a classical computer by an algorithm scaling as $\text{poly}(N)\exp(k)$ [1]. We show that, for a quantum circuit to simulate quantum chaotic behavior, it is both necessary and sufficient that $k = \Theta(N)$. This result implies the impossibility of simulating quantum chaos on a classical computer.

Quantum chaos is QUANTUM

Quantum chaos is a certain type of complex quantum behavior that results in the exponential decay of out-of-time-order correlation functions (OTOC)[2–4] efficient operator spreading[5, 6], small fluctuations of the purity[7] and information scrambling[8, 9]. All these quantities can be unified in a single framework[10] which shows that, in order to simulate quantum chaos, one needs at least a unitary 4-design, that is, a set of unitary operators that reproduces up to the four moments of the Haar distribution over the unitary group $\mathcal{U}(d)$ in an d -dimensional Hilbert space. Here, we define quantum chaos for a quantum evolution in terms of attaining the Haar value for general multi-point OTOC, that is, the value that would be reached by a random unitary operator in $\mathcal{U}(d)$. We consider a subgroup of the unitary group - the Clifford group - which

Conclusions and Outlook

In this paper, we showed that in a random Clifford circuit with N qubits, $\Theta(N)$ non Clifford gates are both necessary and sufficient to simulate quantum chaos. As a consequence, quantum chaos cannot be efficiently simulated on a classical computer, as the cost for simulating such circuits is exponential in the non Clifford resources.

In perspective, there are several open questions. One could generalize many of these results by proving that an ϵ -approximate $2t$ -OTOC characterizes an ϵ -approximate t -design. Although the scaling is fixed to be $\Theta(N)$, the actual number of non Clifford resources is undetermined and it would be of practical importance in obtaining approximate t -designs with a noisy, intermediate-scale quantum computer. One could thus study the optimal arrangement of non Clifford resources. A related question is the onset of irreversibility in a closed quantum system in the sense of entanglement complexity[16] is driven by the doping of a Clifford circuit. Similarly, it would be interesting to show how the entanglement spectrum statistics converges with the doping[11].



Conclusions

- ❑ In “*Scrambling and Lyapunov exponent in spatially extended systems*” work studied the simpler, **2-point** and **integrated OTOC** growth, demonstrating an extended exponential regime in random Clifford circuits as a *toy model* of scrambling.
- ❑ quantum chaos **cannot be efficiently simulated on a classical computer**, as the cost for simulating such circuits is exponential in the non-Clifford resources.



THANK
YOU

